10 Generative models

- throughout most of this course, we have talked about how to make predictions
- in some form or another, we used deep neural networks to learn mappings from data examples to labels
- this kind of learning is called *discriminative learning*, meaning that we would like to be able to discriminate between photos of cats and photos of dogs
- classifiers and regressors are both examples of discriminative learning
- also, neural networks trained by backpropagation have obtained remarkable performance in discriminative learning on large complicated datasets
- for example, classification accuracies on high-resolution images have gone from useless to human-level in just 5-6 years

10 Generative models

- but there is more to machine learning than just solving discriminative tasks
- for example, given a large dataset, without any labels, we might want to learn a model that concisely captures the characteristics of this data
- given such a model, we could sample synthetic data examples that resemble the distribution of the training data
- for example, given a large corpus of photographs of faces, we might want to be able to generate a new photorealistic image that looks like it might plausibly have come from the same dataset
- this kind of learning is called *generative modeling*

10 Generative models

- until recently, we had no method that could synthesize novel photorealistic images
- but the success of deep neural networks for discriminative learning opened up new possibilities
- one big trend over the last years has been the application of discriminative deep networks to overcome challenges in problems that we do not generally think of as supervised learning problems
- the recurrent neural network language models are one example of using a discriminative network (trained to predict the next character) that, once trained, can act as a generative model